

Project Design Phase-II Technology Stack (Architecture & Stack)

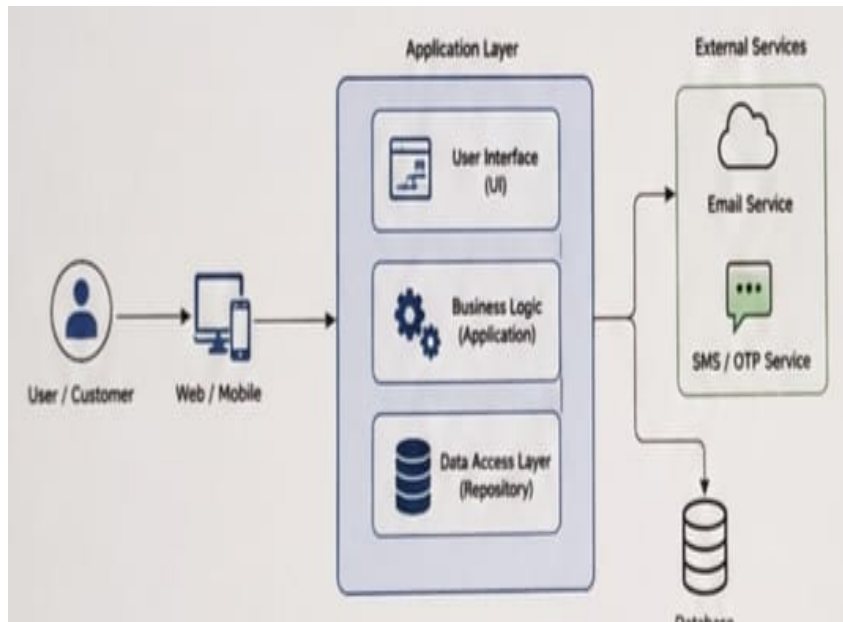
Date	31 January 2026
Team ID	
Project Name	Customer registry
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

1. Include all the processes (As an application logic / Technology Block)
2. Provide infrastructural demarcation (Local / Cloud)
3. Indicate external interfaces (third party APIs etc.)
4. Indicate Data Storage components / services
5. Indicate Interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web & Mobile interface for user registration, login and profile management	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Handle user registration, login, validation, and business rules	Java / Spring Boot
3.	Application Logic-2	Manage emailSMS notifications and OTP verification	java
4.	Application Logic-3	RESTful APIs for communication between client and server	REST API
5.	Database	Store customer and application data	MySQL
6.	Cloud Database	Cloud hosted database service	AWS RDS(MYSQL)
7.	File Storage	Store user uploaded documents and images	AWS S3
8.	External API-1	Email service API for sending verification emails	sendGrid/AWS SES
9.	External API-2	SMS service API for OTP and alerts	Twilio
10.	Machine Learning Model	(Optional) Fraud detection / user behavior analysis	Python(scikit-learn)
11.	Infrastructure (Server / Cloud)	Application deployment and cloud infrastructure Technology	AWS EC2, AWS S3, Docker, Kubernetes

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Spring BOOt

S.No	Characteristics	Description	Technology
2.	Security Implementations	List all the security / access controls implemented (eg. Authentication, Authorization, Encryption)	JWT, BCrypt,HTTPS
3.	Scalable Architecture	Ability to handle growing number of users	Microservices, Load Balancer
4.	Availability	System should be availability 24/7with minimal downtime	AWS Cloud, Auto Scaling
5.	Performance	Fast response time for user requests	Caching (Redis), CDN

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>